

Residue Manifold Learning: Constraint Sampling, Structural Recovery, and Phase-Lock Geometry

Dan Hawkley

Independent Researcher
Boulder, CO

May 4, 2026

Abstract

We study structure, sampling, and representation in modular arithmetic through the mod30 residue manifold. Prime-support residues form an eight-lane discrete manifold embedded in $\mathbb{Z}/30\mathbb{Z}$. We show that constraint-aligned sampling improves signal access, that nonnegative matrix factorization (NMF) recovers the manifold compactly, and that sparse autoencoder (SAE) configurations can fragment or dilute structure despite comparable reconstruction performance. We introduce the Constraint-Guided Coherence Score (CGCS), a bounded metric for structural fidelity, and demonstrate that representation regimes separate into phase-locked, partial, and diluted regions. Finally, we give a geometric interpretation of CGCS as a cosine alignment constraint with a phase-lock boundary near 45° . This connects sampling, representation, and geometry in a unified framework.

1 Introduction

Understanding how structure, sampling, and representation interact is central to modern computational learning. Recent work has shown that apparent computational advantages can arise from access to structured data representations rather than fundamentally different computational power.

We investigate this principle in a controlled modular setting. Prime residues modulo 30 form a discrete, structured manifold. This provides a minimal environment to study:

- Structure: existence of constrained residue lanes
- Sampling: access aligned vs unaligned with structure
- Representation: recovery vs fragmentation of structure

We show that these elements interact to determine structural fidelity.

2 Residue Manifold Structure

Let $n \in \mathbb{Z}$ and consider residues modulo 30:

$$r = n \bmod 30.$$

Prime numbers lie only in the residue set:

$$\{1, 7, 11, 13, 17, 19, 23, 29\}.$$

This defines an eight-lane discrete manifold embedded in $\mathbb{Z}/30\mathbb{Z}$.

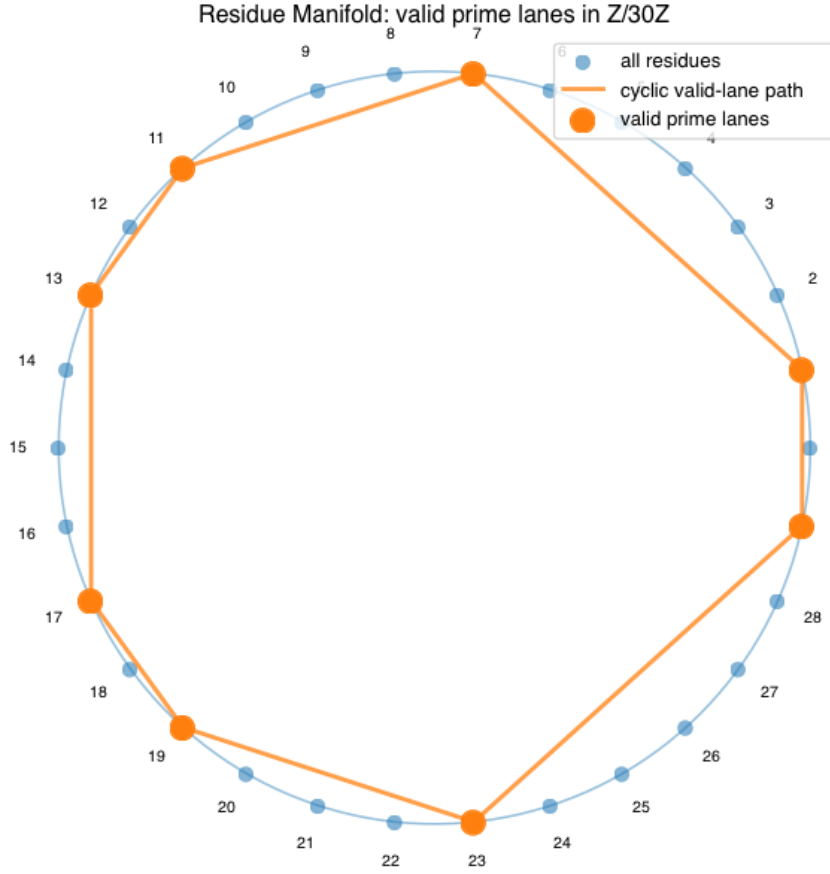


Figure 1: Residue manifold embedded on a circle. Valid prime-support residues form eight discrete angular lanes.

3 Constraint Sampling

We compare two sampling regimes:

- Uniform sampling over all residues
- Constrained sampling restricted to valid lanes

Uniform sampling wastes probability mass outside the manifold:

$$\text{signal rate} \approx \frac{8}{30}.$$

Constrained sampling achieves:

signal rate = 1.

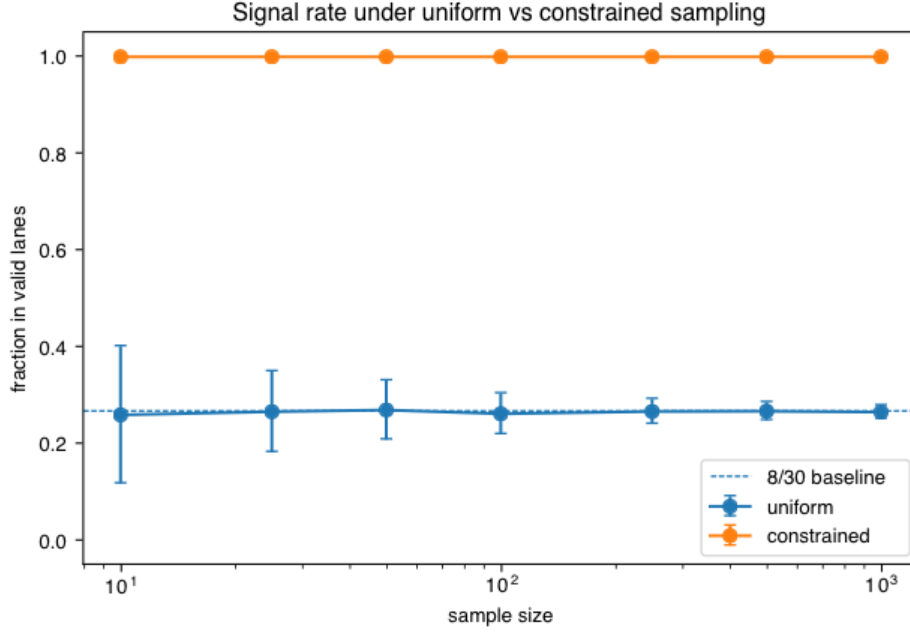


Figure 2: Signal rate under uniform vs constrained sampling.

4 Representation and Recovery

4.1 NMF Recovery

Nonnegative matrix factorization (NMF) recovers the residue manifold as a compact basis:

$$X \approx WH,$$

where H aligns with residue lanes.

4.2 SAE Dilution

Sparse autoencoders (SAE) may:

- duplicate features
- leave features inactive
- spread features across residues

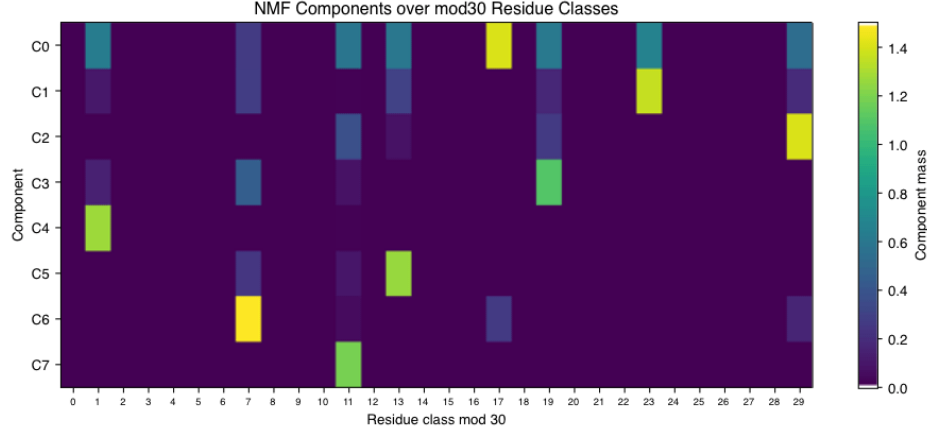


Figure 3: NMF components align with residue lanes.

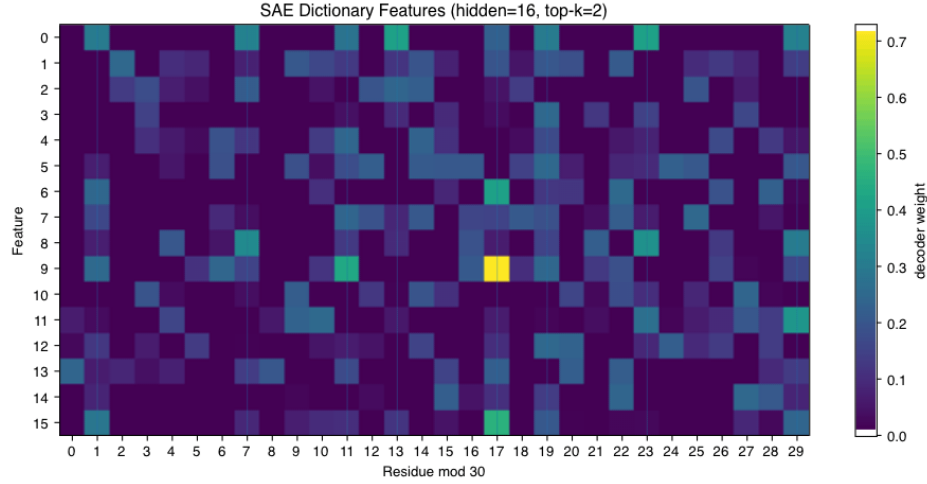


Figure 4: SAE dictionary features showing redundancy and diffusion.

5 Phase Diagram of Structural Regimes

We combine results into a phase diagram over model capacity:

- Coverage of valid lanes
- Alignment to residue structure
- Redundancy and inactive features

We observe distinct regimes:

- Recovered (compact, aligned)
- Partial
- Diluted
- Fragmented

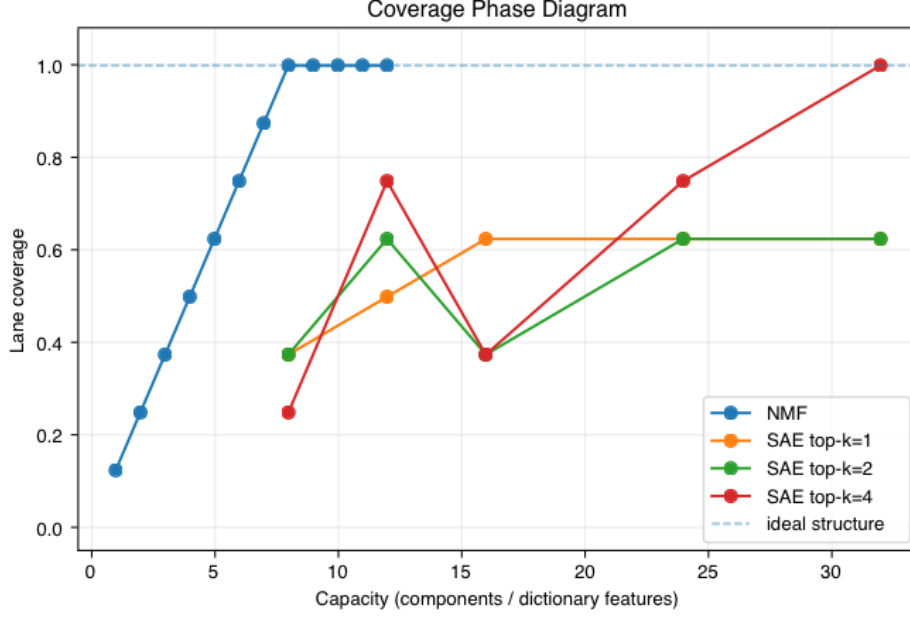


Figure 5: Coverage vs capacity for NMF and SAE.

6 Constraint-Guided Coherence Score (CGCS)

We define:

$$\text{CGCS} = (\text{coverage}) \times (\text{alignment}) \times (\text{redundancy penalty}) \times (\text{dead feature penalty}) \times (\text{reconstruction penalty})$$

CGCS is bounded in $[0, 1]$.

We define a phase-lock threshold:

$$\text{CGCS} \geq \frac{24}{25}.$$

7 Phase-Lock Geometry

We interpret CGCS geometrically using cosine similarity:

$$\cos \theta = \frac{A \cdot B}{\|A\| \|B\|}.$$

Structure quality corresponds to alignment angle θ .

- $\theta \approx 0^\circ$ — fully aligned
- $\theta \leq 45^\circ$ — phase-locked
- $\theta > 45^\circ$ — degraded

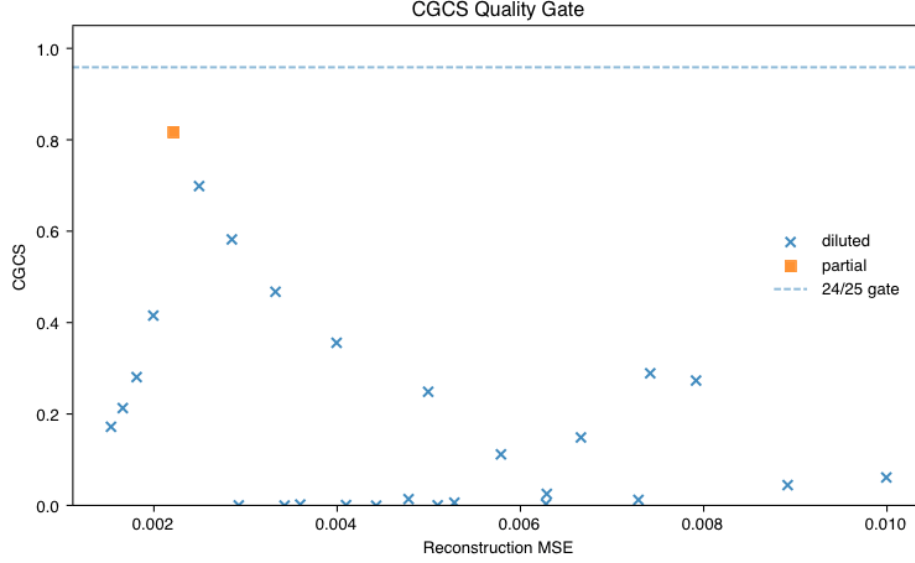


Figure 6: CGCS separates phase-locked and degraded regimes.

8 Discussion

We show that:

- Structure exists independently of representation
- Sampling determines access to structure
- Representation determines whether structure is preserved

Compact factorizations preserve structure, while overparameterized sparse representations may dilute it.

9 Conclusion

Residue-manifold learning provides a minimal setting to study structural fidelity. CGCS unifies coverage, alignment, and representation quality into a single metric. Its geometric interpretation reveals a phase-lock boundary governing structure preservation.

References

- [1] E. Tang, *A quantum-inspired classical algorithm for recommendation systems*, arXiv:1807.04271 (2018).

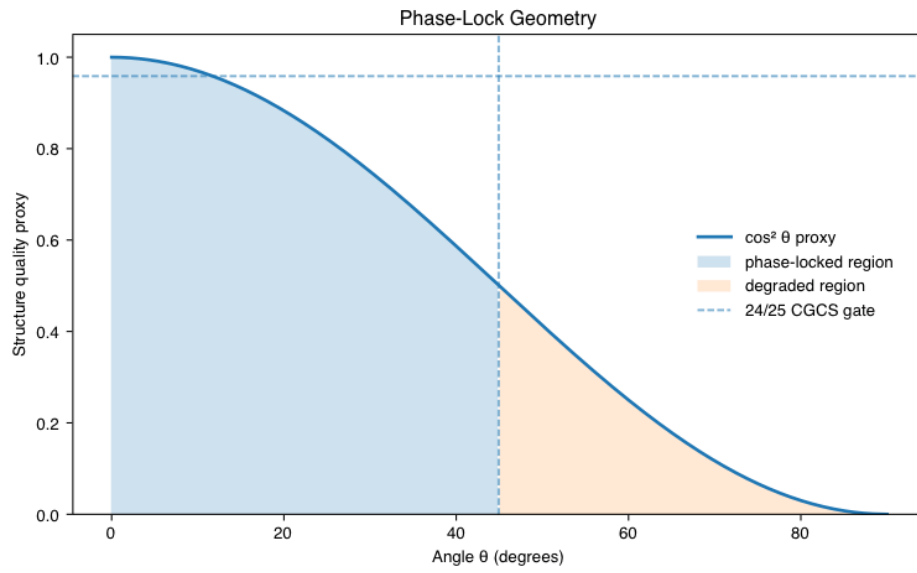


Figure 7: Phase-lock region defined by angular threshold.